

Design Calculation Sheet - VBF-T5R1-CALC-01

VBF-T5R1 delivery package - generated 2026-08-25

Design Calculation Sheet - VBF-T5R1-CALC-01 - sheet 'Inputs'

Parameter	Value	Unit	Formula/Definition	Source
Nominal cell capacity	5	Ah	parameter set	Chen2020/OKane2022 dump
Voltage window	2.5 - 4.2	V	lower/upper cut-off	parameter set
Electrode height	0.065	m	parameter set	dump
Electrode width	1.58	m	parameter set	dump
Electrode area	0.1027	m2	height x width	calc-energy area_m2
Positive electrode thickness	7.56e-05	m	design override	bridge/r7b_negR261.json
Positive electrode porosity	0.335	-	set default (not overridden)	dump
Positive electrode density	3262	kg/m3	set	dump
Positive active material volume fraction	0.665	-	set	dump
Positive particle radius	2.61e-06	m	design override	bridge
Negative electrode thickness	0.0001058	m	design override	bridge
Negative electrode porosity	0.45	-	design override	bridge
Negative electrode density	1657	kg/m3	set	dump
Negative active material volume fraction	0.75	-	set	dump
Negative particle radius	2.61e-06	m	design override (decisive plating lever)	bridge
Separator thickness	8e-06	m	design override	bridge
Separator porosity	0.55	-	design override	bridge
Separator density	397	kg/m3	set	dump
Al collector thickness	8e-06	m	design override	bridge
Cu collector thickness	6e-06	m	design override	bridge
Electrolyte conductivity (target)	2	S/m	design override (upper-bound estimate)	bridge
Electrolyte diffusivity (target)	4.5e-10	m2/s	design override (upper-bound estimate)	bridge
Cation transference number (target)	0.45	-	design override (upper-bound estimate)	bridge
Total heat transfer coefficient	60	W/m2/K	design override (immersion-class)	bridge
Cell cooling surface area	0.00531	m2	set	dump

Parameter	Value	Unit	Formula/Definition	Source
1C protocol ambient	298.15	K	protocol definition	pybamm_runner PROTOCOLS
4C protocol ambient	318.15	K	protocol definition	pybamm_runner PROTOCOLS

Design Calculation Sheet - VBF-T5R1-CALC-01 - sheet 'CapacityEnergy'

Quantity	Chen2020 (DFN)	OKane2022 (DFN)	Unit	Source
1C discharge capacity	6.2104	6.2086	Ah	cell/r7b_*_1c_dfn.json via calc-energy
Rated energy (1C integral)	22.2364	22.2024	Wh	calc-energy energy_wh
Midpoint voltage	3.7978	3.5908	V	calc-energy
DC resistance (calc-energy caliber)	3.5	17.93	mOhm	calc-energy dcr_ohm
Power density	34.75	6.79	kW/kg	calc-energy
4C T_max (45 C ambient)	326.829	328.189	K	cell/r7b_*_4c_dfn.json
4C anode surface potential min	0.01826	0.01798	V	cell/r7b_*_4c_dfn.json (>0 = no plating)
4C charge acceptance (physical)	4.73	5	Ah	charge-segment duration x 20 A / 3600 (library capacity_ah key under-reports 5x - bug, not used)

Design Calculation Sheet - VBF-T5R1-CALC-01 - sheet 'EnergyDensity'

Layer	Thickness m	Porosity	Density kg/m3	kg/m2	g per cell	Share	Source
Positive electrode	7.56e-05	0.335	3262	0.163993788	16.842	48.6%	calc-energy layer_kg_m2
Negative electrode	0.0001058	0.45	1657	0.09642083	9.902	28.6%	calc-energy layer_kg_m2
Positive CC (Al 8um)	8e-06	0	2700	0.0216	2.218	6.4%	calc-energy layer_kg_m2
Negative CC (Cu 6um)	6e-06	0	8960	0.05376	5.521	15.9%	calc-energy layer_kg_m2
Separator	8e-06	0.55	397	0.0014292	0.147	0.4%	calc-energy layer_kg_m2
TOTAL	0.0002034	None	None	0.03463	34.631	100%	calc-energy mass_kg
Energy density (Wh/kg)	None	None	None	None	642.098	E/mass	calc-energy (contract caliber, electrolyte excluded)
Volumetric ED (Wh/L)	None	None	None	None	1064.49	E/volume	calc-energy
Layer stack volume	None	None	None	None	20.89	cm3	calc-energy volume_m3

Design Calculation Sheet - VBF-T5R1-CALC-01 - sheet 'NPMass'

Quantity	Value	Formula / note	Source
N/P ratio (thickness-ratio caliber)	1.4	105.8/75.6 = 1.3995; caliber used in proposals R4C-R6B. Parameter set defines fully-charged initial concentrations but no 0%-SOC stoich keys, so capacity-density form not computable from set keys alone.	propose entries + set dump
N/P empirical support	anode-limited 1C	capacity grew 5.73 -> 6.42 Ah when anode 85.2 -> 110 um (R4C) -> anode limits discharge -> N/P >= 1 consistent	R4C evaluate entries
Electrolyte mass (reference)	9.53 g	pore vol 7.943 cm3 x 1.2 g/cm3 (literature density, annotated)	mechanical (design_spec section 3)
Total mass incl. electrolyte (reference)	44.16 g	34.631 + 9.53	mechanical

Quantity	Value	Formula / note	Source
kg/kWh total (contract caliber)	1.557	mass_kg / E_kWh	calc-energy
kg/kWh total incl. electrolyte	1.986	(mass + electrolyte) / E_kWh	mechanical

Design Calculation Sheet - VBF-T5R1-CALC-01 - sheet 'Process'

Process parameter	Value	Formula	Source
Positive areal density	163.99 g/m ²	$75.6 \times 10^{-6} \times (1 - 0.335) \times 3262$	mechanical
Negative areal density	96.42 g/m ²	$105.8 \times 10^{-6} \times (1 - 0.45) \times 1657$	mechanical
Positive compaction density	2.169 g/cm ³	$3262 \times 0.665 / 1000$	mechanical (divide by 1000!)
Negative compaction density	0.911 g/cm ³	$1657 \times 0.55 / 1000$	mechanical
Electrolyte fill amount	9.53 g (7.943 cm ³)	pore volume x 1.2 g/cm ³ (literature density, annotated)	mechanical
Formation recommendation	0.1C CC to 4.2 V, 25 C, 2 cycles	design recommended value; production-line value requires tuning	design note